

where we have slightly abused notation by allowing the symbol $\mathbf{R}$ to refer to both the involution matrix and the corresponding mapping $\mathbf{z} \mapsto \mathbf{R}\mathbf{z}$.

We may seek numerical methods that mimic this property, which we refer to time-reversible numerical methods. Recall from the previous chapter that a numerical method $\mathcal{G}_h$ was said to be symmetric if it satisfied

$$\mathcal{G}_{-h} = \mathcal{G}_h^{-1}.$$

It does not automatically follow that a symmetric numerical method is time-reversible, i.e. satisfies a relation inspired by (3.14), namely

$$\mathbf{R} \circ \mathcal{G}_h \circ \mathbf{R} \circ \mathcal{G}_h = \text{Id}. \quad (3.15)$$

The additional condition needed to ensure (3.15) is the *affine invariance* property (or invariance with respect to the specific affine transformation $\mathbf{z} \mapsto \mathbf{R}\mathbf{z}$).

Definition 3.1 (Affine Invariance) Let

$$\frac{d\mathbf{z}}{dt} = \mathbf{f}(\mathbf{z}) \quad (3.16)$$

be a given ordinary differential equation. The transformation $\tilde{\mathbf{z}} = \mathbf{A}\mathbf{z}$, where $\mathbf{A}$ is a non-singular square matrix, results in the modified differential equation

$$\frac{d\tilde{\mathbf{z}}}{dt} = \mathbf{A}\mathbf{f}(\mathbf{A}^{-1}\tilde{\mathbf{z}}). \quad (3.17)$$

Let $\mathbf{z}_0, \mathbf{z}_1, \dots$ be a sequence of points obtained by application of the numerical method $\mathcal{G}_h$ to (3.16). If the same method, when applied to (3.17) produces the set of points $\mathbf{A}\mathbf{z}_0, \mathbf{A}\mathbf{z}_1, \dots$, then we say that the numerical method is invariant with respect to the transformation $\tilde{\mathbf{z}} = \mathbf{A}\mathbf{z}$. An affine invariant method is one which is invariant under any such transformation.

If a differential equation is time-reversible with respect to the involution $\mathbf{R}$, and we apply a method which is affine invariant and symmetric, then the method will be time-reversible (3.15).

All Runge-Kutta methods and Partitioned Runge-Kutta methods are affine invariant, thus, if they are also symmetric, then they preserve time-reversal symmetry.

3.6.3 Comparing Symplectic and Time-Reversible Methods

The symplectic property

$$\mathcal{F}_t'^T \mathbf{J} \mathcal{F}_t' = \mathbf{J}$$